Homework 4

1. Problems and applications

1.

a) A cold snap destroys oranges, so the supply of orange juice shifts left. Price rises, quantity falls.

b) Warm weather in New England makes Caribbean vacations less attractive, so demand for Caribbean hotel rooms shifts left. Price falls, quantity falls.

c) War disrupts oil supply, so the supply of gasoline shifts left and the price of gasoline rises. Gasoline and Cadillacs are complements, so demand for used Cadillacs shifts left and their price fdls.

3.

a) Tastes change: demand increases. Price ↑, quantity ↑.

b) Input price (steel) rises: supply decreases. Price ↑, quantity ↓.

c) Better technology: supply increases. Price ↓, quantity ↑.

d) Price of a substitute (SUVs) rises: demand increases. Price ↑, quantity ↑.

e) Lower wealth (minivans are a normal good): demand decreases. Price ↓, quantity ↓.

6.

a) Cotton, an input, becomes more expensive: supply decreases. Price ↑, quantity ↓.

b) Leather jackets, a substitute, get cheaper: demand decreases. Price ↓, quantity ↓.

c) Required morning exercise: demand increases. Price ↑, quantity ↑.

d) New knitting machines (technology): supply increases. Price ↓, quantity ↑.

9.

The health news increases demand (D shifts right), and the fertilizer increases supply (S shifts right). Equilibrium quantity definitely rises. The effect on price is ambiguous: it rises if the demand shift is larger and falls if the supply shift is larger.

11.

a) The supply curve is vertical (perfectly inelastic) at 8,000 tickets, because the arena has a fixed number of seats no matter the price.

b) Equilibrium: price $8, quantity 8,000 tickets.

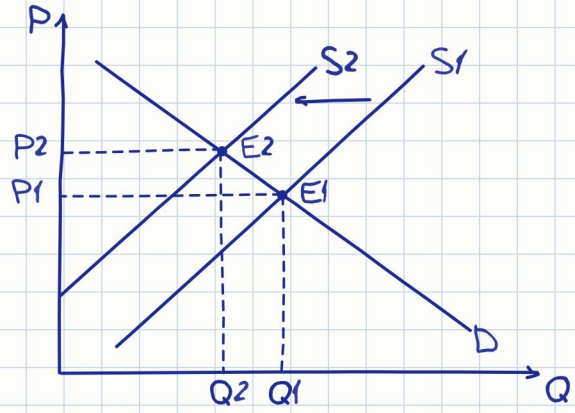
2. Workbook

1-a, 2-b, 3-d, 4-b, 5-c, 6-c, 7-d, 8-b, 9-a, 10-b, 11-a, 12-c, 13-a, 14-c, 15-a, 16-a, 17-c, 18-b, 19-c, 20-c

3. Short essay

a) Main cause: supply

Natural gas is a key input for producing electricity. When Russia invaded Ukraine in February 2022 and cut gas and oil deliveries to Europe, the cost of producing electricity rose sharply. At the same time, the EU's commitment to close traditional thermoelectric and nuclear plants reduced the number of producers and the available capacity. Both factors decreased the supply of electricity, while demand stayed roughly the same.

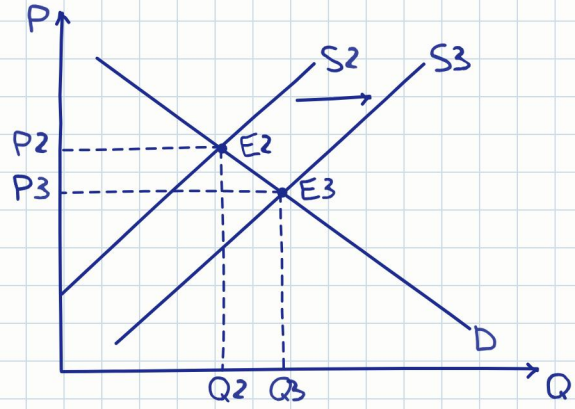


Graph: price on the vertical axis, quantity on the horizontal axis. The supply curve shifts left from S1 to S2. The equilibrium moves from E1 (Q1, P1) to E2 with a higher price P2 and a lower quantity Q2.

b) Advice to the Prime Minister: increase supply in the short run.

The fastest way to lower the price is to shift the supply curve back to the right:

- postpone the closure of existing coal and nuclear plants and keep them running at full capacity;
- import more electricity and gas from alternative suppliers;
- cut taxes on energy production, which lowers producers' costs.



Graph: supply shifts right from S2 to S3. The equilibrium moves from E2 to E3, so the price falls from P2 to P3 and the quantity rises from Q2 to Q3.

A supporting measure is to reduce demand through energy-saving

campaigns. This shifts D to the left and also lowers the price.

The government should avoid a strict price cap below the equilibrium price. A cap lowers the price on paper, but at that price quantity demanded exceeds quantity supplied, which creates a shortage (possible blackouts) and gives producers no incentive to produce more.